PATTERN / TYPE : NEET	ANSWER KEY	PRE-MEDICAL 2022

TOPIC : THERMAL PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A.	1	1	1	2	1	2	2	4	2	3	3	3	3	2	1
Q.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	4	1	4	2	2	4	1	4	2	3	2	4	3	3
Q.	31	32	33	34	35	36	37	38	39	40					
A.	4	2	1	2	2	1	2	4	4	3					

1701CMD302120019	*Your Target is to secure Good Rank in Pre-Medical 2022*	SIKAR

HINT & SOLUTION

1. ANS – 1

Time period $T = 2\pi\sqrt{\frac{l}{g}}$

$\frac{\Delta T}{T} = \frac{1}{2}\frac{\Delta l}{l} = \frac{1}{2}\alpha\Delta\theta$

$= \frac{1}{2} \times 12 \times 10^{-6} \times (40 - 20) = 12 \times 10^{-5}$

$\Delta T = T \times 12 \times 10^{-5}$

$= 24 \times 60 \times 60 \times 12 \times 10^{-5}$

$= 10.3\ s\ day^{-1}$

2. ANS – 1

Energy gained by water (in 1 s)

= energy supplied – energy lost

= (1000 J – 160 J) = 840 J

Total heat required to raise the temperature of water from 27° to 77°

Q = msΔθ.

Hence, the required time,

$t = \frac{ms\Delta\theta}{840} = \frac{2 \times 4200 \times 50}{840}$

$= 500s = 8\,\text{min}\,20\ sec$

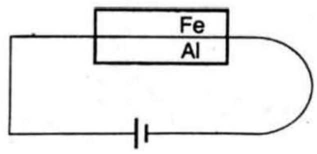
3. ANS – 1

A good absorber is also a good emitter. So black body absorbs all the light but also emits whereas black holes do not emit 99% of radiations.

4. ANS – 2

Since the curved surface of the conductor is thermally insulated, in steady state, the rate of flow of heat at every section will be the same. Hence, the curve between H and x will be a straight line parallel to x axis.

5. ANS – 1



Because $\alpha_{Fe} < \alpha_{Ae}$, Strip bends upward due to heating.

6. ANS – 2

Given - S = 2Cal/cm²/min

But as 1Cal = 4.18J, 1cm = 10^{-2}m and 1 min = 60s

$$S = \frac{2 \times 4.18J}{(10^{-2}m)^2(60s)}$$

≈ 1.4 × 10^3 J/m²s   or S= 1.4 kWm^{-2}

7. ANS – 2

If *f* is the degree of freedom of a gas molecule, then

$$\frac{C_p}{C_v} = \left(\frac{f+2}{f}\right)$$

$$\left(\frac{C_p}{C_v}\right)_A = \frac{29}{22} = \left(\frac{f_A+2}{f_A}\right)$$

f_A=6

$$\left(\frac{C_p}{C_v}\right)_B = \frac{30}{21} = \left(\frac{f_B+2}{f_B}\right)$$

f_B = 5, So *A* has one vibrational degree of freedom and *B* has none

8. ANS – 4

Kinetic energy of a gas molecule

$$E = \frac{3}{2}kT \Rightarrow \therefore E \propto T$$

$$\frac{E_1}{E_2} = \frac{T_1}{T_2} \Rightarrow \frac{E}{E/2} = \frac{300}{T_2}$$

$$T_2 = 150k = 150 - 273 = -123°\text{C}$$

9. ANS – 2

When pressure is increased by 10%

$$P_1 = P\left(1 + \frac{1}{100}\right) = \frac{11}{10}P$$

Then pressure is decreased by 10%

$$P_2 = P_1\left(1 - \frac{1}{100}\right) = \frac{9}{10}P_1$$

$$P_2 = \frac{9}{10}\left(\frac{11}{10}P\right) = \frac{99}{100}P$$

Then temperature is constant

$$PV = P_1V_1 = P_2V_2$$

$$PV = \frac{99}{100}PV_2$$

$$V_2 = \frac{100}{99}V$$

10. ANS – 3

Black seat is good radiator of heat energy therefore on a clear night the black seat of a bicycle becomes wet.

11. ANS – 3

$$Pressure, \quad P = \frac{\rho RT}{M}$$

Hence at constant pressure,

$$\rho T = constant$$

$$\frac{\rho_1}{\rho_2} = \frac{T_1}{T_2}$$

$$\frac{24}{\rho_2} = \frac{273+127}{273+27} = \frac{400}{300}$$

$$\rho_2 = 18\ units$$

12. ANS – 3

$$T_1 = 127°C, \qquad T_2 = 400K$$

From Stefan's law

$$\frac{\frac{dQ_1}{dt}}{\frac{dQ_2}{dt}} = \frac{T_1^4}{T_2^4} \Rightarrow \left(\frac{1}{16}\right)^{\frac{1}{4}} = \frac{T_1}{T_2}$$

$$\frac{1}{2} = \frac{400}{T_2} \Rightarrow T_2 = 800\ K$$

$$\Rightarrow T_2 = 800 - 273 = 527\,°C$$

13. ANS – 3

Because Planck's law explains the distribution of energy correctly at low temperature as well as at high temperature.

14. ANS – 2

$\Delta Q = \Delta U + \Delta W$

$Given, \Delta Q = 200J \ and \ \Delta W = -100J$

$\Delta U = \Delta Q - \Delta W$

$\Delta U = 200 - (-100) = 300J$

15. ANS – 1

$Q_2 = 2000$ cal, as$COP = \dfrac{Q_2}{W}$

$W = \dfrac{COP}{Q_2} = 500$ cal

$W = 500 \times 4.2 = 2100$ J

16. ANS – 2

$$\frac{H_1}{H_2} = \frac{m_1c_1}{m_2c_2}$$

$$\frac{H_1}{H_2} = \frac{\frac{4}{3}\pi r_1{}^3\rho_1 C_1}{\frac{4}{3}\pi r_2{}^3\rho_2 C_2}$$

$$\frac{H_1}{H_2} = \frac{r_1{}^3}{r_2{}^3} \times \frac{\rho_1}{\rho_2} \times \frac{C_1}{C_2}$$

$$\frac{H_1}{H_2} = \frac{1^3}{2^3} \times \frac{2}{1} \times \frac{1}{3} = \frac{1}{12}$$

17. ANS – 4

Heat required for $100\,g$ ice to melt is

$Q_1 = mL = 100(80) = 8000\ cal$

Heat given by $300g$ water when it cools to $0°C$

$Q_2 = ms\Delta\theta = 300(1)(25-0)$

$Q_2 = 7500\ cal$

As $Q_2 < Q_1$ Therefore ice cannot melt completely and final temperature of mixture is $0°C$.

18. ANS – 1

It is given that $\frac{K_1}{K_2} = \frac{1}{3}$

Let $K_1 = K$ & $K_2 = 3K$

The temperature in the junction in contact

$$\theta = \frac{K_1\theta_1 + K_2\theta_1}{K_1 + K_2}$$

$$\theta = \frac{1 \times 100 + 3 \times 100}{1+3} = 25°C$$

19. ANS – 4

Thermal equilibrium implies that the temperatures of gases are same and molecules are also same therefore

$P_aV_a = P_bV_b$

20. ANS – 2

Given that amount of heat given is equal to the decrease in internal energy,

dQ = - dU

$C = -C_V$

$$\Rightarrow \frac{R}{\gamma-1} + \frac{R}{1-x} = -\frac{R}{\gamma-1}$$

$$\Rightarrow \frac{R}{\gamma-1} + \frac{R}{1-\frac{6}{5}} = -\frac{R}{\gamma-1}$$

$\Rightarrow \frac{5}{2} = \frac{1}{\gamma-1} \ \Rightarrow \gamma - 1 = \frac{2}{5} \ \Rightarrow \gamma = \frac{7}{5}$ The gas may be diatomic or polyatomic linear.

21.ANS – 2

$$\text{Absorption Power} = \frac{\text{Heat absorbed}}{\text{total heat given}} = \frac{q}{p}$$

22. ANS – 4

Relation between °C *and* °F

$$\frac{C}{5} = \frac{F - 32}{9}$$

$$\frac{C}{5} = \frac{F - 32}{9}$$

$$\frac{C}{5} = \frac{140 - 32}{9}$$

$$C = 60°\text{C}$$

23. ANS – 1

During free expansion of an ideal gas no work is done and also no heat is supplied from outside. Therefore from first law of thermodynamics no change in internal energy.

Q = W = 0 and ΔU=O

24.ANS – 4

Total internal energy of system

$$U = U_{\text{oxygen}} + U_{\text{argon}}$$

$$U = n_1 \frac{f_1}{2} RT + n_2 \frac{f_2}{2} RT$$

Degree of freedom is $f_1 = 5$

(for oxygen) and $f_2 = 3$ (for argon)

$$U = 2\frac{5}{2}RT + 4\frac{3}{2}RT$$

$$U = 5RT + 6RT$$

$$U = 11RT$$

25. ANS – 2

When radius is decreased by dr

Decrease in surface energy = Heat required for vaporization

$$(8\pi r dr) \times T = 4\pi r^2 \text{dr } \rho L \Rightarrow r = \frac{2T}{\rho L}$$

26. ANS – 3

$$\gamma = 1 + \frac{2}{f} = 1 + \frac{2}{6} = \frac{4}{3}$$

Root mean square speed, $c = \sqrt{\frac{3p}{\rho}}$ and speed of sound, $v = \sqrt{\frac{\gamma p}{\rho}}$

$$\therefore \frac{v}{c} = \sqrt{\frac{\gamma}{3}} \Rightarrow \frac{v}{c} = \sqrt{\frac{4}{3}} \Rightarrow v = \frac{2c}{3}$$

27. ANS – 2

Rate of cooling,

$$\frac{d\theta}{dt} = -k(\theta - \theta_0)$$

From 40°C to 36°C,

$$\theta_1 = \frac{40 + 36}{2} = 38°\text{C} \Rightarrow \left(\frac{d\theta}{dt}\right)_1 = \frac{4}{2} = 2°\text{C/min}$$

From 36°C to 32°C,

$$\theta_2 = \frac{36 + 32}{2} = 34°\text{C} \Rightarrow \left(\frac{d\theta}{dt}\right)_2 = \frac{4}{t}$$

$$\frac{\left(\frac{d\theta}{dt}\right)_1}{\left(\frac{d\theta}{dt}\right)_2} = \frac{k(\theta_1 - \theta_0)}{k(\theta_2 - \theta_0)} \Rightarrow \frac{\frac{4}{2}}{\frac{4}{t}} = \frac{38 - 20}{34 - 20}$$

$$\Rightarrow t = 2\ min\ 33s$$

28. ANS – 4

$$Area = Work\ done = W = \frac{1}{2} \times 2V \times 3P = 3PV$$

29. ANS – 3

Infinite thermal capacity implies that there would be practically no change in temperature whether heat is taken in or given out.

30. ANS – 3

The efficiency of the engine is

$$\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{375}{500} = 0.25$$

Heat consumed per cycle Q_1 = 600 kcal

Now, as, $\left[\eta = \frac{Q_1 - Q_2}{Q_1}\right]$

Heat rejected to the sink is

$$Q_2 = Q_1 - \eta . Q_1 = 600 - 0.25 \times 600 = 450\, kcal$$

31. ANS – 4

During adiabatic expansion we know $TV^{\gamma-1} = Constant$

OR $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$

For a monoatomic gas, $\gamma = \frac{5}{3}$

$$\therefore \frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1} = \left(\frac{AL_2}{AL_1}\right)^{\frac{5}{3}-1} = \left(\frac{L_2}{L_1}\right)^{\frac{2}{3}}$$

(Where A is the cross section area of the piston)

32. ANS – 2

Molar heat capacity = Molar mass x Specific heat capacity.So, the molar heat capacity at constant pressure and constant volume will be 28 C_P and 28 C_V respectively

$$\therefore 28C_P - 28C_V = R \;\; or \;\; C_P - C_V = \frac{R}{28}$$

33. ANS – 1

Rate of emission of energy $= \sigma T^4 s$

Let m_1 be the mass of sphere, C is specific heat and $\left(\frac{d\theta}{dt}\right)$, The rate of cooling

$$For\ sphere,\ \ \sigma T^4 s = m_1 C \left(\frac{d\theta}{dt}\right)_s \ldots\ldots\ldots\ldots(i)$$

Let m_2 be the mass of cube, C is specific heat and $\left(\frac{d\theta}{dt}\right)$, The rate of cooling

$$For\ cube\ \ \sigma T^4 s = m_2 C \left(\frac{d\theta}{dt}\right)_c \ldots\ldots\ldots\ldots\ldots\ldots(ii)$$

$from\ equation\ (i) and\ (ii)$

$$\frac{\left(\frac{d\theta}{dt}\right)_s}{\left(\frac{d\theta}{dt}\right)_c} = \frac{m_2}{m_1} = \frac{a^3\rho}{\frac{4}{3}\pi r^3 \rho}$$

Where a is the side of cube and r is the radius of sphere and ρ is the density.

$$Required\ Ratio,\ \ \frac{R_s}{R_C} = \frac{3a^3}{4\pi r^3}$$

But since S(Surface area) is the same,

$$6a^2 = 4\pi r^2 \ or\ a^2 = \left(\frac{2}{3}\right)\pi r^2$$

$$\therefore \frac{R_s}{R_C} = \frac{3\left(\frac{2\pi r^2}{3}\right)^{\frac{3}{2}}}{4\pi r^3} = \frac{2\pi\sqrt{2\pi}}{\sqrt{3}(4\pi)} = \sqrt{\frac{2\pi}{12}} = \sqrt{\frac{\pi}{6}}$$

34. ANS – 2

If mass of a gas is constant then there is variation in density of gas with the variation of temperature and pressure because the molar volume of the gas goes on changing because

$$\rho = \frac{M}{V}$$

We know from ideal gas law $P = \frac{\rho RT}{M}$

For point A,

$$P = \frac{\rho_0 RT_0}{M} \Rightarrow M = \frac{\rho_0 RT_0}{p} - - - 1$$

For point B

$$3P = \frac{\rho R(2T_0)}{M} = \frac{\rho R(2T_0)}{\rho_0 RT_0} \times p \;\; [\text{By eq. (i)}]$$

$$\rho = \frac{3\rho_0}{2}$$

35. ANS – 2

Given PV^{-2} = constant

Comparing with PV^N = constant, we get $N = -2$

$$C = C_v + \frac{R}{1-N} = \frac{5R}{2} + \frac{R}{1+2}$$

$$= \frac{5R}{2} + \frac{R}{3} = \frac{15R + 2R}{6} = \frac{17R}{6}$$

36. ANS – 1

According to Wein's displacement law,

$\lambda_m \propto \frac{1}{T}$ or $\frac{c}{v_m} \propto \frac{1}{T}$

$v_m \propto T$

Hence curve A represents the correct variation.

37. ANS – 2

Temperature at state P=T_0, since P lies on the isotherm of temperature T_0.if T be the temperature at Q, then for the adiabatic process B, we have

$$T_0 V_0^{\gamma-1} = T(2V_0)^{\gamma-1}$$

$$T = \frac{T}{2^{\gamma-1}} = \frac{T_0}{2^{\frac{2}{3}}}$$

Change in internal energy of the gas is

$$\Delta U = C_V(T - T_0)$$

$$= \left(\frac{3R}{2}\right)\left(\frac{T_0}{2^{\frac{2}{3}}} - T_0\right) = \frac{3RT_0\left(1 - 2^{\frac{2}{3}}\right)}{2 \times 2^{\frac{2}{3}}} = -4.6T_0$$

38. ANS – 4

Initial kinetic energy of the system,

$$U_i = \frac{5}{2}NRT$$

Final kinetic energy of the system,

$$U_f = 2n\frac{3}{2}RT + \frac{5}{2}(N-n)RT = \frac{1}{2}nRT + \frac{5}{2}NRT$$

$$\Delta U = Q = \frac{1}{2}nRT$$

39. ANS – 4

According to Wien's displacement Law

$$\lambda_m T = b$$

$$T = \frac{b}{\lambda_m} = \frac{2892 \times 10^{-6}}{14.46 \times 10^{-6}} = 200K$$

40. ANS – 3

Internal energy does not depend on path; it only depends on initial and final states